

19.2 An incorrect covariance matrix

Suppose the mean is right but the errors are correlated or have unequal variances. Write

$$E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}, \quad \text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{V}, \quad (19.2.1)$$

with $\mathbf{V}$ positive definite and $\mathbf{X}$ of full rank $p < n$. The analyst fits ordinary least squares and reports $s^2(\mathbf{X}^\top \mathbf{X})^{-1}$ as if $\mathbf{V} = \mathbf{I}$. Two things go wrong: $\hat{\boldsymbol{\beta}}$ is no longer the best linear unbiased estimator (Corollary 7.2.3), and $s^2(\mathbf{X}^\top \mathbf{X})^{-1}$ no longer estimates its covariance. The second failure is usually much more serious.

19.2.1 What happens to the estimator and its standard error

The bounds below need a fact about traces of projections, a special case of an inequality of Ky Fan.

Lemma 19.2.1 — Trace of a compressed matrix

Let $\mathbf{A}$ be symmetric with eigenvalues $\lambda_1 \geq \dots \geq \lambda_n$, and let $\mathbf{P}$ be a symmetric idempotent matrix of rank k . Then

$$\sum_{i=n-k+1}^n \lambda_i \leq \text{tr}(\mathbf{P}\mathbf{A}) \leq \sum_{i=1}^k \lambda_i.$$

Proof. By Theorem 1.7.2, $\mathbf{A} = \sum_i \lambda_i \mathbf{q}_i \mathbf{q}_i^\top$ with orthonormal $\mathbf{q}_i$. So $\text{tr}(\mathbf{P}\mathbf{A}) = \sum_i \lambda_i w_i$ with $w_i = \mathbf{q}_i^\top \mathbf{P} \mathbf{q}_i = \|\mathbf{P} \mathbf{q}_i\|^2$. Each w_i lies in $[0, 1]$ because the eigenvalues of $\mathbf{P}$ are 0 or 1 (Proposition 6.3.4), so $0 \leq \mathbf{q}_i^\top \mathbf{P} \mathbf{q}_i \leq \mathbf{q}_i^\top \mathbf{q}_i = 1$, and $\sum_i w_i = \text{tr}(\mathbf{P} \sum_i \mathbf{q}_i \mathbf{q}_i^\top) = \text{tr} \mathbf{P} = k$. Then

$$\begin{aligned} \sum_i \lambda_i w_i - \sum_{i \leq k} \lambda_i &= \sum_{i \leq k} \lambda_i (w_i - 1) + \sum_{i > k} \lambda_i w_i \\ &\leq \lambda_k \sum_{i \leq k} (w_i - 1) + \lambda_k \sum_{i > k} w_i = \lambda_k \left(\sum_i w_i - k \right) = 0, \end{aligned}$$

since $w_i - 1 \leq 0$ and $\lambda_i \geq \lambda_k$ for $i \leq k$, while $w_i \geq 0$ and $\lambda_i \leq \lambda_k$ for $i > k$. The lower bound follows by applying this to $-\mathbf{A}$. ■

Theorem 19.2.2 — Least squares under a wrong covariance

Under Eq. 19.2.1, let $\mathbf{a} \in \mathbb{R}^p$ be nonzero and let $v = s^2 \mathbf{a}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{a}$ be the usual estimate of $\text{Var}(\mathbf{a}^\top \hat{\boldsymbol{\beta}})$.

$$(a) \ E(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta} \text{ and } \text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{V} \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1}.$$

- (b) $E(s^2) = \sigma^2 \operatorname{tr}((\mathbf{I} - \mathbf{M})\mathbf{V})/(n - p)$.
 (c) With $\mathbf{u} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{a} / \|\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{a}\|$, a unit vector in $\mathcal{C}(\mathbf{X})$,

$$\frac{E(v)}{\operatorname{Var}(\mathbf{a}^\top \hat{\boldsymbol{\beta}})} = \frac{\operatorname{tr}((\mathbf{I} - \mathbf{M})\mathbf{V})/(n - p)}{\mathbf{u}^\top \mathbf{V} \mathbf{u}}. \quad (19.2.2)$$

- (d) Let $\lambda_1 \geq \dots \geq \lambda_n > 0$ be the eigenvalues of $\mathbf{V}$, let $\bar{\lambda}_-$ be the mean of the $n - p$ smallest and $\bar{\lambda}_+$ the mean of the $n - p$ largest. Then

$$\frac{\bar{\lambda}_-}{\lambda_1} \leq \frac{E(v)}{\operatorname{Var}(\mathbf{a}^\top \hat{\boldsymbol{\beta}})} \leq \frac{\bar{\lambda}_+}{\lambda_n}.$$

Proof.

- (a) is Proposition 5.5.7(b) with $\boldsymbol{\Sigma} = \sigma^2 \mathbf{V}$. (b) is Theorem 2.3.1 with the mean term vanishing, since $(\mathbf{I} - \mathbf{M})\mathbf{X}\boldsymbol{\beta} = \mathbf{0}$. (c) Put $\mathbf{c} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{a}$. By (a), $\operatorname{Var}(\mathbf{a}^\top \hat{\boldsymbol{\beta}}) = \sigma^2 \mathbf{c}^\top \mathbf{V} \mathbf{c}$, and $\mathbf{c}^\top \mathbf{c} = \mathbf{a}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{a}$. By (b), $E(v) = \sigma^2 \mathbf{c}^\top \mathbf{c} \operatorname{tr}((\mathbf{I} - \mathbf{M})\mathbf{V})/(n - p)$. Divide, and write $\mathbf{u} = \mathbf{c}/\|\mathbf{c}\|$.
 (b) $\mathbf{I} - \mathbf{M}$ is symmetric idempotent of rank $n - p$, so by Lemma 19.2.1 the numerator of Eq. 19.2.2 lies in $[\bar{\lambda}_-, \bar{\lambda}_+]$. By Theorem 1.7.7 the denominator lies in $[\lambda_n, \lambda_1]$. ■

Part (c) shows what the usual standard error gets wrong. The numerator of Eq. 19.2.2 averages $\mathbf{V}$ over the residual space, which is what s^2 measures. The denominator evaluates $\mathbf{V}$ in the one direction $\mathbf{u}$ that $\mathbf{a}^\top \hat{\boldsymbol{\beta}}$ uses. When the errors vary more along $\mathbf{u}$ than on average, the usual standard error is too small. The bounds in (d), due to Swindel (1968), hold for every design and linear function and can be attained (Exercise (C1)). They are typically wide.

19.2.2 When ordinary least squares loses little

Generalized least squares has variance $\sigma^2 \mathbf{a}^\top (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{a}$ for $\mathbf{a}^\top \boldsymbol{\beta}$ (Corollary 7.2.3). How much larger can the variance of $\mathbf{a}^\top \hat{\boldsymbol{\beta}}$ be? The answer again depends only on the extreme eigenvalues of $\mathbf{V}$.

Theorem 19.2.3 — Efficiency of least squares

Under Eq. 19.2.1, with λ_1 and λ_n the largest and smallest eigenvalues of $\mathbf{V}$,

$$\frac{\mathbf{a}^\top (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{a}}{\mathbf{a}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{V} \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{a}} \geq \frac{4\lambda_1 \lambda_n}{(\lambda_1 + \lambda_n)^2}$$

for every $\mathbf{a} \neq \mathbf{0}$. The bound is attained for $p = 1$ and $\mathbf{X} = \mathbf{x} = (\mathbf{q}_1 + \mathbf{q}_n)/\sqrt{2}$, where $\mathbf{q}_1, \mathbf{q}_n$ are unit eigenvectors for λ_1, λ_n .

Proof. Write $\mathbf{X} = \mathbf{Q}\mathbf{R}$ with $\mathbf{Q}$ having orthonormal columns and $\mathbf{R}$ nonsingular (Theorem 1.5.2), and put $\mathbf{b} = \mathbf{R}^{-\top} \mathbf{a}$, $\mathbf{A} = \mathbf{Q}^\top \mathbf{V} \mathbf{Q}$ and $\mathbf{B} = \mathbf{Q}^\top \mathbf{V}^{-1} \mathbf{Q}$, both positive definite. Then $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{V} \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} = \mathbf{R}^{-1} \mathbf{A} \mathbf{R}^{-\top}$ and $(\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} = \mathbf{R}^{-1} \mathbf{B}^{-1} \mathbf{R}^{-\top}$, so the ratio is $\mathbf{b}^\top \mathbf{B}^{-1} \mathbf{b} / \mathbf{b}^\top \mathbf{A} \mathbf{b}$. It suffices to show $\mathbf{A} \leq \kappa \mathbf{B}^{-1}$ in the nonnegative definite

order, with $\kappa = (\lambda_1 + \lambda_n)^2 / (4\lambda_1\lambda_n)$.

Every eigenvalue λ of $\mathbf{V}$ satisfies $(\lambda - \lambda_1)(\lambda - \lambda_n) \leq 0$, that is, $\lambda + \lambda_1\lambda_n/\lambda \leq \lambda_1 + \lambda_n$. By the spectral decomposition this says $\mathbf{V} + \lambda_1\lambda_n\mathbf{V}^{-1} \leq (\lambda_1 + \lambda_n)\mathbf{I}$. Multiplying by $\mathbf{Q}^\top$ on the left and $\mathbf{Q}$ on the right preserves the order, so

$$\mathbf{A} \leq (\lambda_1 + \lambda_n)\mathbf{I} - \lambda_1\lambda_n\mathbf{B}.$$

Now let $\mu > 0$ be any eigenvalue of $\mathbf{B}$. By the inequality between arithmetic and geometric means, $\kappa/\mu + \lambda_1\lambda_n\mu \geq 2\sqrt{\kappa\lambda_1\lambda_n} = \lambda_1 + \lambda_n$. Applied along the eigenvectors of $\mathbf{B}$, this gives $(\lambda_1 + \lambda_n)\mathbf{I} - \lambda_1\lambda_n\mathbf{B} \leq \kappa\mathbf{B}^{-1}$, and hence $\mathbf{A} \leq \kappa\mathbf{B}^{-1}$.

For the attainment, with $\mathbf{x} = (\mathbf{q}_1 + \mathbf{q}_n)/\sqrt{2}$ we have $\mathbf{x}^\top \mathbf{x} = 1$, $\mathbf{x}^\top \mathbf{V} \mathbf{x} = (\lambda_1 + \lambda_n)/2$ and $\mathbf{x}^\top \mathbf{V}^{-1} \mathbf{x} = (1/\lambda_1 + 1/\lambda_n)/2$. The ratio is $1/((\mathbf{x}^\top \mathbf{V} \mathbf{x})(\mathbf{x}^\top \mathbf{V}^{-1} \mathbf{x})) = 4\lambda_1\lambda_n/(\lambda_1 + \lambda_n)^2$. ■

The scalar inequality behind the proof, $(\mathbf{x}^\top \mathbf{V} \mathbf{x})(\mathbf{x}^\top \mathbf{V}^{-1} \mathbf{x}) \leq \kappa(\mathbf{x}^\top \mathbf{x})^2$, is Kantorovich's inequality, and the proof above is a matrix form of it. Bloomfield and Watson (1975) and Knott (1975) found the corresponding sharp bound when efficiency is measured by the determinant of the whole covariance matrix.

The bound is a worst case over all designs. For particular designs ordinary least squares can be fully efficient. Kruskal's theorem (Theorem 6.9.3) says that $\hat{\boldsymbol{\beta}}$ equals the generalized least squares estimate for every $\mathbf{Y}$ iff $\mathcal{C}(\mathbf{V}\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$, and then the efficiency is one. Even then the standard errors may be wrong.

Example. (Equicorrelated errors)

Let $\mathbf{V} = (1 - \rho)\mathbf{I} + \rho\mathbf{1}\mathbf{1}^\top$, with $-1/(n - 1) < \rho < 1$ so that $\mathbf{V}$ is positive definite, and let the model contain an intercept, $\mathbf{X} = [\mathbf{1}, \mathbf{X}_1]$. It arises when all observations share one random shock. Since $\mathbf{V}\mathbf{X} = (1 - \rho)\mathbf{X} + \rho\mathbf{1}(\mathbf{1}^\top \mathbf{X})$ has columns in $\mathcal{C}(\mathbf{X})$, Kruskal's condition holds and $\hat{\boldsymbol{\beta}}$ is the best linear unbiased estimator. Its covariance is not $\sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$, however. Because $\mathbf{X}\mathbf{e}_1 = \mathbf{1}$ for the first standard basis vector $\mathbf{e}_1$, we have $(\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{X}^\top \mathbf{1} = \mathbf{e}_1$, and Theorem 19.2.2(a) gives

$$\text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2(1 - \rho)(\mathbf{X}^\top \mathbf{X})^{-1} + \sigma^2\rho\mathbf{e}_1\mathbf{e}_1^\top.$$

By Exercise (B2), $E(s^2) = \sigma^2(1 - \rho)$. So $s^2(\mathbf{X}^\top \mathbf{X})^{-1}$ is exactly unbiased for the covariance of the slopes, whose information comes from comparisons in which the common shock cancels. For the intercept it misses the term $\sigma^2\rho$. When $\rho > 0$ this term does not shrink as n grows, so the intercept is not even consistent, and its reported standard error goes to zero anyway (Exercise (B2)). The same holds for any fitted mean $\mathbf{x}_0^\top \hat{\boldsymbol{\beta}}$, since $\mathbf{x}_0^\top \mathbf{e}_1 = 1$ whenever $\mathbf{x}_0$ has a leading 1.

19.2.3 Autocorrelated errors

Example. (A trend with autocorrelated errors)

Take $n = 20$ equally spaced times and errors with $v_{st} = \rho^{|s-t|}$, the covariance of a stationary first-order autoregression, with $\rho = 0.6$. Fit a straight line in two designs, a

linear trend $x_t = t$ and an alternating regressor $x_t = (-1)^t$. For the trend, the expected usual variance of the slope is 0.256 times its true variance, so the root of the expected usual variance is 0.506 of the true standard error. Yet ordinary least squares is nearly efficient: generalized least squares has 0.905 of its variance. For the alternating regressor the error goes the other way. The ratio is 3.271, so the standard error is too large by a factor 1.809, and the efficiency is 0.950. Swindel's bounds for this $\mathbf{V}$ are 0.193 and 4.31, so both designs lie well towards the extremes. The Kantorovich bound, 0.2351 from $\lambda_1 = 3.757$ and $\lambda_n = 0.2514$, is far below either efficiency.

In 100000 simulated series with normal errors, the nominal 95% t interval for the slope covered the true value in 0.693 of them for the trend and 0.9976 for the alternating regressor. Figure 19.2.1 shows the whole range of ρ .

```
import numpy as np
n = 20
t = np.arange(1, n + 1)
designs = {"trend": t.astype(float), "alternating": (-1.0) ** t}

def ar1(rho):
    return rho ** np.abs(t[:, None] - t[None, :])

def slope_summary(x, V):
    """True variance, mean of the usual estimate, GLS variance of the slope (sigma =
    1)."""
    X = np.column_stack([np.ones(n), x])
    G = np.linalg.inv(X.T @ X)
    A = G @ X.T                                # beta_hat = A y
    true_var = (A @ V @ A.T)[1, 1]              # the sandwich
    M = X @ A
    Es2 = np.trace((np.eye(n) - M) @ V) / (n - 2) # E(s^2)
    usual = Es2 * G[1, 1]                        # E of s^2 [(X'X)^{-1}]_{11}
    gls = np.linalg.inv(X.T @ np.linalg.solve(V, X))[1, 1]
    return true_var, usual, gls

rho = 0.6
for name, x in designs.items():
    true_var, usual, gls = slope_summary(x, ar1(rho))
    print(f"{name:12s} usual/true = {usual / true_var:.3f}    GLS/OLS variance = {gls
    / true_var:.3f}")
```

[Run this code online](#)

```
from scipy import stats
rng = np.random.default_rng(1902)
reps = 100_000
L = np.linalg.cholesky(ar1(rho))
E = rng.normal(size=(reps, n)) @ L.T          # rows are AR(1) error vectors
tq = stats.t.ppf(0.975, n - 2)
for name, x in designs.items():
    X = np.column_stack([np.ones(n), x])
    G = np.linalg.inv(X.T @ X)
    B = E @ (G @ X.T).T                        # beta_hat - beta, one row per
    data set
    s2 = np.sum((E - B @ X.T) ** 2, axis=1) / (n - 2)
    cover = np.mean(np.abs(B[:, 1]) <= tq * np.sqrt(s2 * G[1, 1]))
    print(f"{name:12s} coverage of the nominal 95% interval: {cover:.4f}")
```

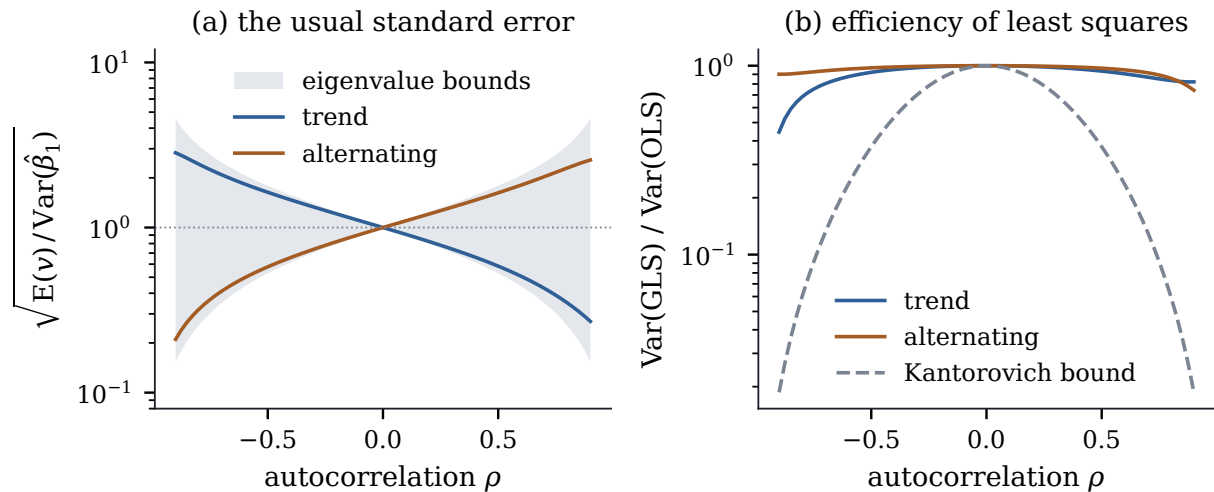


Figure 19.2.1: Straight-line regression with $n = 20$ and first-order autoregressive errors. (a) The square root of the ratio of the expected usual variance of the slope to its true variance, for a trend and an alternating regressor, with Swindel's bounds (shaded). (b) The efficiency of ordinary least squares relative to generalized least squares, with the Kantorovich bound of Theorem 19.2.3.

[Run this code online](#)

Eq. 19.2.2 explains the pattern. A smooth regressor puts $\mathbf{u}$ along slowly varying directions, where positively correlated errors have their largest variance; an alternating one puts it where positive correlation cancels. Most time-series regressors are smooth, which is why positive autocorrelation usually makes ordinary standard errors too small.

Idea. A wrong covariance matrix is mainly a problem for *standard errors*, not *estimates*: $s^2(\mathbf{X}^\top \mathbf{X})^{-1}$ can be wrong by a large factor in either direction, and nothing in the fitted model reveals this. Chapter 21 shows how to estimate the correct covariance Eq. 5.5.3 without knowing $\mathbf{V}$, and how to detect serial correlation. Chapter 31 treats generalized least squares when $\mathbf{V}$ is known up to a few parameters (Definition 31.4.1).

19.2.4 Exercises

19.2.4.1 A. Check your understanding

Exercise A1.

Let $\mathbf{V} = \text{diag}(v_1, \dots, v_n)$. Show that $E(s^2) = \sigma^2 \sum_i (1 - h_{ii}) v_i / (n - p)$, a weighted average of the variances $\sigma^2 v_i$.

Exercise A2.

Show that the bounds of Theorem 19.2.2(d) do not change when $\mathbf{V}$ is replaced by $c\mathbf{V}$ for $c > 0$, and that both equal 1 iff $\mathbf{V}$ is a multiple of $\mathbf{I}$.

19.2.4.2 B. Practice

Exercise B1.

For the regression through the origin $E(Y_i) = \beta x_i$ with $\text{Var}(Y_i) = \sigma^2 v_i$, show that $\text{Var}(\hat{\beta}) = \sigma^2 \sum_i h_i v_i / \sum_i x_i^2$ and $E(v) = \sigma^2 \sum_i (1 - h_i) v_i / ((n - 1) \sum_i x_i^2)$, where $h_i = x_i^2 / \sum_j x_j^2$ are the leverages. Conclude that the usual standard error is too small on average iff the leverage-weighted average of the v_i exceeds the $(1 - h_i)$ -weighted average.

Solution. $\hat{\beta} = \sum_i x_i Y_i / S$ with $S = \sum_j x_j^2$, so $\text{Var}(\hat{\beta}) = \sigma^2 \sum_i x_i^2 v_i / S^2 = \sigma^2 \sum_i h_i v_i / S$. [Exercise \(A1\)](#) with $p = 1$ gives $E(s^2)$, and $v = s^2 / S$. The weights h_i sum to 1, and so do $(1 - h_i) / (n - 1)$. The usual variance is too small on average iff $\sum_i h_i v_i > \sum_i (1 - h_i) v_i / (n - 1)$. In particular, when the error variance grows with $|x_i|$, the points that dominate $\hat{\beta}$ are the noisiest, and the usual standard error understates the truth.

Exercise B2.

In [Example \(Equicorrelated errors\)](#) with $\rho > 0$, show that $\text{Var}(\hat{\beta}_0) \geq \sigma^2 \rho$ for every n , while $E(v) \rightarrow 0$ for the intercept when $[(\mathbf{X}^\top \mathbf{X})^{-1}]_{11} \rightarrow 0$. Explain the result by writing $\varepsilon_i = \sqrt{\rho} \sigma Z + \sqrt{1 - \rho} \sigma U_i$ with independent standardized $Z, U_1, \dots, U_n$.

Solution. From the covariance in the example, $\text{Var}(\hat{\beta}_0) = \sigma^2 (1 - \rho) [(\mathbf{X}^\top \mathbf{X})^{-1}]_{11} + \sigma^2 \rho \geq \sigma^2 \rho$, while $E(v) = \sigma^2 (1 - \rho) [(\mathbf{X}^\top \mathbf{X})^{-1}]_{11}$. The representation has covariance $\sigma^2 \rho \mathbf{1}\mathbf{1}^\top + \sigma^2 (1 - \rho) \mathbf{I} = \sigma^2 \mathbf{V}$. The shared term $\sqrt{\rho} \sigma Z \mathbf{1}$ lies in $\mathcal{C}(\mathbf{X})$ and is absorbed entirely by the intercept, since $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{1} = \mathbf{e}_1$. It never averages out, and it leaves no trace in the residuals, so s^2 cannot see it.

19.2.4.3 C. Going deeper**Exercise C1.**

Show that both bounds of [Theorem 19.2.2\(d\)](#) are attained. *Hint:* take $\mathcal{C}(\mathbf{X})$ spanned by eigenvectors of $\mathbf{V}$, and choose $\mathbf{a}$ so that $\mathbf{u}$ is one of them.

Solution. Let $\mathbf{q}_1, \dots, \mathbf{q}_n$ be orthonormal eigenvectors for $\lambda_1 \geq \dots \geq \lambda_n$. If $\mathbf{X} = [\mathbf{q}_1, \dots, \mathbf{q}_p]$, then $\mathbf{I} - \mathbf{M} = \sum_{i>p} \mathbf{q}_i \mathbf{q}_i^\top$ and $\text{tr}((\mathbf{I} - \mathbf{M})\mathbf{V}) = \sum_{i>p} \lambda_i = (n - p) \bar{\lambda}_-$. With $\mathbf{a} = \mathbf{e}_1$, $\mathbf{u} = \mathbf{q}_1$ and $\mathbf{u}^\top \mathbf{V} \mathbf{u} = \lambda_1$, so the ratio [Eq. 19.2.2](#) equals $\bar{\lambda}_- / \lambda_1$. Symmetrically, $\mathbf{X} = [\mathbf{q}_{n-p+1}, \dots, \mathbf{q}_n]$ and $\mathbf{u} = \mathbf{q}_n$ give $\bar{\lambda}_+ / \lambda_n$.